

# Beyond the Clock: Rethinking Time in Technologies for People with Dementia

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## ABSTRACT

We explore the intricate interplay between humans, time, and technology, highlighting the prevalence of objective temporal perspectives, such as the clock in modern technologies. Our focus narrows on assistive technologies, where the prevailing adherence to seemingly objective clock time sidelines subjective human experiences, which is a critical oversight, especially for vulnerable groups like people with dementia. This case underscores the evident temporal misfit between lived experiences and standardized clock time imposed by technology. Consequently, assistive technologies inadequately address nuanced temporal challenges in daily life. Through dementia-driven scenarios and temporal misfits with assistive technology, we advocate for a more inclusive and intuitive technology approach, urging a reconsideration of the pervasive use of clock time in HCI. We urge the community to embrace multi-temporal lenses that can accommodate a range of temporal perspectives and move beyond the technological imposition of commoditized clock time.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

## KEYWORDS

Dementia, time, technology, temporality, temporal-misfit

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## 1 INTRODUCTION

### Story of Alice

*Once upon a time, not far from Wonderland, lived a gentle soul named Alice. However, Alice was not traversing through a wonderland of enchanting tea parties; instead, she was confronted with dementia, a condition that required her to navigate the intricate corridors within her own mind— where time had different rules.*

*At the start, the clocks on walls and tables tick-tocked in unison chiming a familiar rhythm. Technology beeped with notifications as the screens glowed. The old and new artifacts created a symphony of familiarity. However, as days passed, time in Wonderland began to twist and turn, much like a mischievous rabbit zigzagging through a maze. The clocks that once chimed to a known rhythm now ticked at their own whims, playing hide-and-seek with the steady beat of reality. Appointments and dates, like curious creatures, seemed to slip away into the rabbit hole of Forgetfulness.*

*Technology, once a reliable guide through time, transformed into a curious puzzle for Alice. Notifications, like enchanting riddles, left Alice pondering over their meaning. The glowing screens of technology became a portal to a reality where past, present, and future intertwined in a kaleidoscope of confusion. As dementia progressed, time turned into whimsical echoes, and glowing screens became magical mirrors reflecting fragments of her life. The rabbit hole of Forgetfulness stretched endlessly, and gadgets transformed into curious artifacts from a distant past.*

*Yet, amid the chaos, Alice found moments of clarity. In familiar gazes and gentle touches. In them, she discovered fragments of reality that transcended the Wonderland of her mind. The progression of dementia may have distorted the artifacts that showed the time of the clock, but the essence of her connection with the world remained. In Alice's mind, where time had its own rules, and technology that sometimes played the role of a mysterious guide, moments passed by— a testament to the resilience of the human spirit in the face of a challenging reality.<sup>1</sup>*

Time-related challenges are among the earliest reported changes in dementia and these issues get worse as the condition advances [85]. About 55 million people worldwide currently live with dementia, and 10 million new cases are estimated to be recorded each year [64]. While there is no cure available, technology is emerging as a helping hand in dementia for promoting safety, addressing communication and memory loss, providing multi-sensory stimulation, and acting as a memory enhancer [15]. Previous research on dementia has explored temporal aspects of the condition [38, 68, 74], but their translation into assistive technology design remains limited. Current technology for dementia is largely designed for the measurable aspect of time [12, 27, 87], thus not addressing flexible, subjective experiences and cultural needs.

The quantifiable part of time is mostly related to the clock, which precisely measures time through money and timekeeping as well as more subjectively through culture and biology, and together, these factors encourage individual action through social structures and morals [82]. For temporal artifacts to effectively convey time to humans, they need to share a sense of rhythm and temporality with humans [56]. Over time, humans and artifacts have come to share a sense of rhythm that has moved from fluid norms captured through changes in nature to technology's fixed clock time [3]. Existing technology is made to work with little or rather no regard for the diverse temporal experiences of humans [54]. HCI and design scholars also have questioned the need to rethink time in the design field [70, 71, 73], but not enough has been done to look beyond time as a fixed, linear narrative. This amounts to a *temporal misfit* between humans and technologies, as the modern-day, digital timekeepers do not coincide with the different facets of human temporalities, more so in the case of dementia as seen through the story of Alice.

The functionalist perspective of time has become the norm in HCI research [73]. To go back, *timekeeping* started with the need to coordinate and organize activities in human settlements [3]. Primarily concerned with measurement, time evolved into a commodity during industrialization, emphasizing its efficient use [53]. In the digital age, clocks are seamlessly integrated into technologies, subtly yet significantly influencing people's daily routines [56]. *Cultural time* encapsulates the diverse perspectives on time that persist alongside unified global norms. This diversity is reflected in how people from different cultures communicate about time through gestures, spatial references, and metaphors [26, 72]. Despite the global adoption of the clock, time perception in cultures varies on a scale of linear to flexible time, thus significantly influencing the way people from different cultures manage time and plan activities of daily living [59]. *Biological time* is that of the individual human experience, including the perception of time passage [41], the internal sense of time [7], and how daily experiences over time impact overall well-being [90]. Bringing together these diverse temporal notions is crucial to cater to the nuanced temporal challenges faced by people with dementia, for whom the misfit with time and technologies is more explicit.

In this paper, we first briefly look at the landscape of assistive technologies for dementia. Then, we visualize the experience of time in dementia and the mismatch between technology and people through two scenarios. We briefly dive into the evolution of time-keeping and time of various cultures to look at the lost perspectives of time and make the modern-day dependency on clocks explicit. Finally, the paper nudges the CHI community to re-think the future of technology design through a multi-temporal lens that is inclusive of the flexible and intuitive perspectives of time to look beyond the technological impositions of commoditized time.

## 2 ASSISTIVE TECHNOLOGY FOR DEMENTIA

Dementia is an umbrella term for various progressive brain conditions, leading to a decline in memory, reasoning, communication, and other challenges [64]. As dementia progresses, individuals experience worsening conditions that impact their daily time management, affecting autonomy and increasing dependency on caregivers [45, 68]. In response to the growing challenges posed by the condition, assistive technologies are being proposed as a promising tool for empowering the lives of those affected.

Assistive technologies designed for individuals with dementia can be broadly categorized based on their intended use, including diagnosis, assessment, care delivery, and support for aging at home [6]. Particularly, technologies designed for delivering care and supporting aging at home are designed for different purposes like providing social support [22, 28, 35], companionship [19, 20], navigation [13], reminiscence [42, 86] and assistance in daily tasks [79]. Assistive technology for time communicates essential temporal information, such as day-night transitions, day, date, and time [27, 81, 87]. Occupational therapists recommend assistive technologies for time to improve time orientation, reduce caregiver dependency, enhance the sense of control, and increase overall independence among individuals with dementia [67].

Despite the promising role of assistive technologies, barriers to their widespread acceptance and use persist. Addressing these barriers is crucial to ensure that individuals with dementia can fully benefit from these technological advancements. Existing reviews extensively cover assistive technologies, evaluating their effectiveness and limitations [62, 66, 77, 91]. Reported barriers for people with dementia include declining health conditions, low motivation, perceived lack of need, limited technology experience, readiness to accept technology, customization options, maintenance demands, cost, and the physical size of the technology [46, 67, 75]. Technologies designed for care delivery and home support need seamless integration with the capabilities and needs of people with dementia, considering their active and diverse engagement with technology. Despite the emphasis on user-centered methodologies and novel approaches like 'warm technology' [44], assistive technologies often fail to align with people's temporal experiences. Human caregivers who are trained to adopt a person-centered care approach can deal with the evolving changes over time [57]. Unlike human caregivers who can be temporally flexible, these technologies predominantly rely on a linear, clock-based notion of time. This may result in technologies not being fully aware of the temporal world experienced by individuals with dementia, leading to responses that do not resonate with their fragmented and shifting temporal realities

<sup>1</sup>The story of Alice is a work of fiction featuring a character inspired by Alice in Wonderland and is co-written with ChatGPT. The story incorporates scenarios influenced by the authors' interactions with people with dementia and their caregivers.

Care pathways in dementia are complex and given the deteriorating condition, the unique needs of every individual and the available resources at hand should work together. People diagnosed with dementia are not passive users of technology; they are an active and diverse user group who face particular restrictions on their technology use. We claim that this is not because of their declining condition, but because of designers', technologists', and researchers' failure to extensively imagine their needs, especially regarding their experience of time. The next section will foreground the mismatch between the temporalities of people diagnosed with dementia and assistive technologies designed for them.

### 3 FOREGROUNDING THE NUANCED TEMPORALITIES OF DEMENTIA

To bridge the gap between technology and the nuanced temporal needs of people with dementia, this section first brings to light the misfit with existing technologies that often go unnoticed and then dives into understanding the unique temporalities of dementia. Foregrounding the temporalities of dementia and technology misfits is done with illustrative scenarios. This prompts the readers to reconsider how assistive technologies can be designed to align with the varied temporal experiences. By acknowledging the nonlinear nature of time in dementia, we aim to open avenues for technology that is not only functional but also deeply attuned to the temporal realities of those it aims to assist.

#### 3.1 Scenarios highlighting the misfit of time

The two scenarios provide a perspective on dementia's changing rhythms and the mismatch with various forms of technology. The scenarios are a result of the authors' observations made in care settings, and conversations with researchers, technologists, designers, and those diagnosed with dementia and their partners. Each scenario has 2 parts: *Paragraph 1* presents a specific context from the daily life of a person with dementia and their interactions with technology. *Paragraph 2* uses an example of a hypothetical book, 'Book of Life' to help empathize with people with dementia and imagine through a metaphor how it could feel to experience the context mentioned in *Paragraph 1*. This book is a collection of all moments of one's life spanning the past, present, and future.

**3.1.1 Scenario: Navigating a temporal whirlwind.** The scenario illustrates people's fragmented experiences such as encountering moments from different times in the past and future while living in the now. People like David may lose their sense of time-dependent action in the now when they are mentally traveling to a past or future moment. For example, responding to the body language of a dog wanting attention is an activity fully taking place in the now. However, remembering if the dog needs to eat is challenging. A temporal awareness is required to choose to feed the dog and thus this care activity may be conducted too often or too little.

David prefers to wait for the partner rather than to eat lunch as the system requested. Or, it could be that he has forgotten and moved on in time in his mind. Thus, technology solely designed based on clock time can be a cause of distress by being loud, startling, and out of sync with a person's experience in situ. The reminders interrupt David's 'now' and it disrupts his already disjointed sense of time. A sudden call to action from a voice robot is simply not

enough. The voice robot needs more context to functionally engage with the person and follow up on action points. Thus, a scheduling technology that is designed to prompt a message at a pre-set clock time fails as it does not share the same notion of now with people with dementia. Conventional technologies struggle to adapt to the fluidity of time perception, leading to a disconnect between the needs of people with dementia and the assistance provided by these technologies.

#### Scenario: Navigating a temporal whirlwind

*It is Monday, at noon in David's home. David is petting his dog while he lingers on memories from the past. A voice calls out, "Hey David! It's time for lunch"! David is startled but he doesn't know where the voice came from. That is his new care robot, a voice assistant that reminds him of his to-dos at a pre-set clock time. But David is starting to get worried. He looks around and doesn't see his wife. Panic sets in. While looking around, he finds a note by the door from his wife that says, "Don't worry, I'll be back soon!". That makes him a bit relaxed. He goes to his favourite spot by the window and looks outside waiting for his wife to arrive.*

*David sees pages from his own 'Book of Life' swirling in a whirlwind around him. He reaches out and catches one, feeling the texture of a moment from his teenage years at the farm, the sun-kissed fields, and laughter echoing in the air. Another page, and he is suddenly transported to the upcoming Christmas festivities, twinkling lights, and the warmth of holiday cheer. There, on another page, are details of tonight's dinner, the mouthwatering aroma, and the clinking of cutlery.*



Figure 1: Navigating a temporal whirlwind

### Scenario: Flowing through time

Lisa has Alzheimer's disease, a form of dementia, and she is living at home with her partner. To keep her connected to the 'now', Lisa has a special watch that she keeps on her coffee table. The watch constantly displays the day, date, and time. The watch also has a feature that shows her images of her special moments. But often, Lisa seems to experience time at a different pace and she feels that time is slipping away too quickly. Like earlier today when the calendar showed an image of her birthday from her college years. Lisa starts talking about the day and mentions that it was celebrated last month. The date on the top right of Lisa's watch is no longer a valid indicator of calculating time passage. For Lisa, it is almost like she blinked, and quickly went ahead in time from her earlier birthday to today.

Lisa goes through her own 'Book of Life' while she is on a boat. She has barely turned three pages, browsing through her college graduation day. But, when she looks up, the boat has sailed far and has landed in a whole new place, the holiday in Hawaii with her parents and siblings. Now, in another part of the boat ride, she read other pages from the book, friends from college whose names she does not entirely recall, singing 'Happy birthday'! She looks up again, and guess what? The boat hasn't gone very far this time even though she read the same number of pages!



Figure 2: Flowing through time

**3.1.2 Scenario: Flowing through time.** In the scenario, reading the 'Book of Life' is an analogy for the uniform clock time, with time passage in dementia as exemplified by a boat on a branched-out river. It is assumed that Lisa is immersed when reading the book. The first time Lisa looks up, she has only traveled down one corner, and the other time her boat has crossed many places and is in a new environment entirely. In dementia, this experience is like that of the present moment stretching into eternity when sitting in the boat, while weeks and months flow by rapidly, disappearing around the corner.

Moments from one's lived experiences act as location markers on the temporal landscape. They not only help people remember aspects of their now as a unique story but also act as anchors in the now. In the above context, technology makes use of images of birthdays. However, there is an opportunity to re-think the calendar time of using these lived moments as fixed signposts on the temporal landscape. Does it even matter to Lisa that it has been a specific number of years since that favorite birthday party of her youth? Maybe not, as long as she can locate and orient herself on the timeline of her lifetime. Technology projects a strict clock time and executes the given role of sending reminders. It assumes that there is a correct or an ideal time for the person to return to specific memories. The clock time in technology is to bring everyone into that singular, definitive perception of time. Thus, technology can override a person's navigation of the temporal landscape in the hope of delivering a 'perfect' memory at the 'right' time.

## 3.2 Unwrapping the temporal dimension of dementia

The experience of time in dementia is described as that of living in a "timeless world" [31]. Edwards discusses the experience of timelessness as recognizing one's existence in space and time with a decline in orientation, indicating a weakened connection with reality [31]. Dementia impacts both the objective aspect related to representing (e.g. clocks) and measuring (e.g. hours) time and the subjective aspect that intricately connects to human experience. In this section, we look at the influence of dementia on both these facets of time.

**3.2.1 Objective time.** In healthcare, time-related problems are separated into body and activity domains [63]. The body domain includes task-based awareness of duration and the ability to know when to start an activity, for how long, and to allocate time for various tasks that are bodily performed. The activity domain looks at the ability to plan and execute daily activities in a specific order, within designated time limits, and adapt to unforeseen events. Nested under these domains, every aspect relates to a measurable aspect of time i.e., clock time. In line with this, a slightly dated (although common) assessment tool for early screening of dementia is the Mini Mental State Examination (MMSE) [50, 89], which includes a Clock Drawing Test [5]. In the MMSE, an understanding and orientation of objective time (clock time) are used to estimate the level of cognitive impairment using the question "What is the (year) (season) (date) (day) (month)?"; this shows that the relation between dementia diagnosis and a distorted sense of time is indeed recognized in healthcare.

We live in a world driven by the clock. Thus, as part of the world, people diagnosed with dementia frequently encounter situations where the world around them moves with the clock, whether they are at home, in care facilities, or in hospitals. People with dementia are likely to feel rushed as they strive to match their natural pace with that of their environment, where everyone is using the clock as a shared reference. However, people with dementia struggle with the use and understanding of a clock. The Clock Drawing Test reveals what might go wrong in relating time to the *physical* representation of a clock; Figure 3 shows examples of possible results in the order of severity. This is one indicator of why we

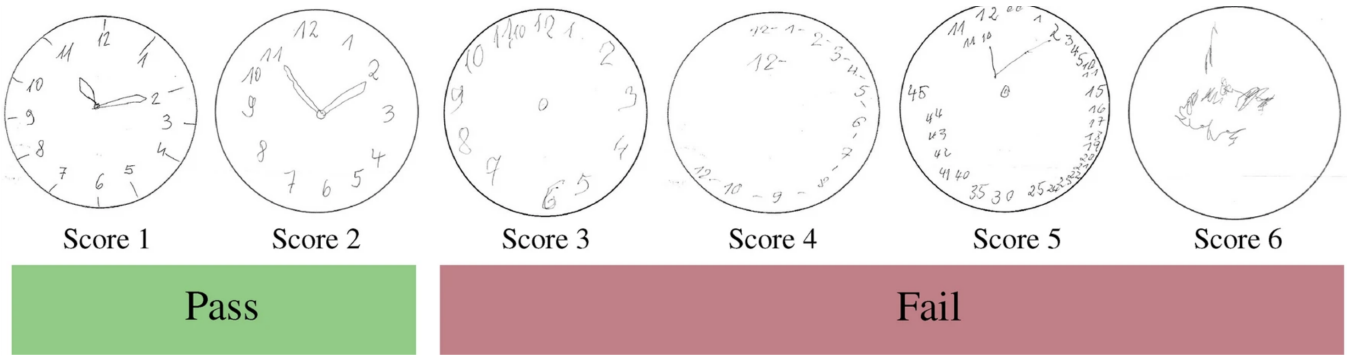


Figure 3: Example of Clock Drawing Test results [21] (adapted under creative commons licence for visual reference)

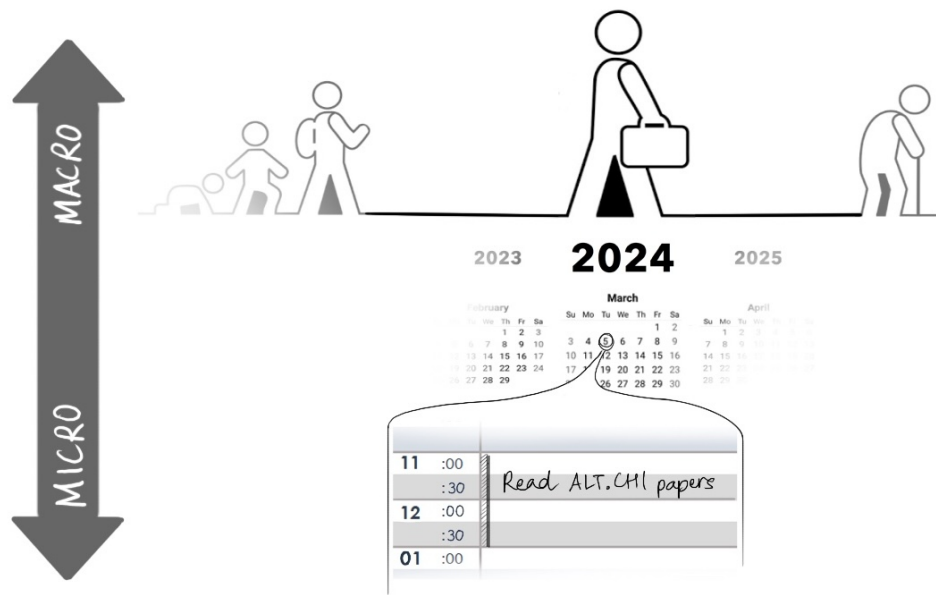


Figure 4: Time as lived experience from micro and macro perspectives (by the first author).

need to look beyond using the clock time as the only temporal reference in dementia.

**3.2.2 Subjective time.** Unlike objective time which is seen as an absolute entity that progresses uniformly with each passing second, subjective time results from an individual’s experience in and with time and this relates to the perception of the now. The momentary impact may seem insignificant when compared across a lifetime but it plays a key role in positioning individuals in the ‘now’- a narrative that is critical for human identity and well-being [25, 69]. Time animates the way individuals experience moments frame by frame at a microscopic level e.g., reading the story of Alice at the start of the paper. These frames arranged chronologically build a coherent story at the macro-level e.g., of a lifetime. This micro-to-macro perspective is illustrated in Figure 4.

The work of Kitwood has been fundamental in prioritizing the subjective psychological needs of people with dementia [47]. Adding

to this, subjective time combines an individual’s past, present, and future, shaping how they perceive and evaluate their temporal experiences across their lifespan [37]. This subjective temporal construct is an intricate interaction between attention, working memory, and episodic memory [2]. The research underlines the close connection between time processing and memory, both of which are centered in the hippocampus—a region in the brain affected by dementia [32]. Due to dementia, the distortion of time and the decline of memory complicate individuals’ navigation of their subjective temporal dimensions. This hinders individuals from adapting to everyday goal-directed behavior and also affects long-term, future-oriented behavior adaptation; both are dependent on subjective time [83]. Also, dementia can take away an individual’s ability to recognize the current moment and the self in relation to others and perform collective action as a result of confabulations [65]. Therefore, dementia significantly impacts engagement with

subjective time, introducing challenges in individuals' sense of self in addition to their ability to plan and navigate across different times.

While the minds of people diagnosed with dementia undergo a change in experiencing time, the world of others is still synchronized by a central clock time. This includes home settings with family, people from their social circles, and healthcare settings with formal caregivers and medical professionals. Technology for dementia, as we saw before, primarily addresses time in an objective sense by making people fit into the time dictated by the clock. This is not only the case for dementia but for all humans; the problem is just more pronounced in the case of dementia.

## 4 TIME THROUGH ARTEFACTS

The exploration of time through temporal concepts and artifacts spans across the domains of physics, timekeeping, culture, and biology [34]. In physics, time traditionally has been given a linear one-way direction- that of an arrow pointing from past to future [18]. Also, timelines presenting a linear story of human evolution have conventionally progressed from left to right [26] and this tendency is also seen in Figure 4. Alongside human evolution, the innovation in timekeeping artifacts encapsulates a history of scientific discoveries and exceptional craftsmanship [78]. But, time is dynamically captured via societal norms, practices, and various aspects of daily life. This section brings forward the diverse perspectives that got lost in this evolution of time management, which contributes to the misfit of temporality in today's assistive technologies.

### 4.1 Technological imposition of time

#### Scenario: Technological Imposition of Time.

*07:00 AM - Alarm rings, and I'm up! Work starts at 9:00 AM. Seems like I have plenty of time, right? Not quite. I just checked the Maps App and trains are running late this morning. That gives me only 50 minutes instead of the usual hour and a half. Panic sets in—I might not make it on time!*

*07:55 AM - Oops, missed the train! Now, I have to reschedule the meeting for 9:00 AM since I can't be there on time! Amidst the bustling train station, I tap on my digital calendar, visually scanning for the next best time slot available. I found one at 2:00 PM and moved the meeting over there. I quickly update the calendar invite for my colleagues!*

*07:57 AM - Ding! A message from my colleague, "Change accepted". Phew! No more rush. I can relax, grab a coffee at the station, and make it back in time for the next train.*

In the scenario 'Technological imposition of time', everything worked out well eventually but we have all probably experienced moments when it has not. The scenario portrays how we, in our networked society, manage our day-to-day life around central time and negotiate time continuously. The domino effect that comes with, for example, proposing an alternative time for a meeting is not only felt at an individual level but ripples into the lives of others within

work and social networks. Managing time goes beyond simply rescheduling a meeting; time management has become a learned skill that requires a person to foresee, meticulously calculate, and then plan time slots to be filled by activities both in private and work life.

The prevalent linear and structured approach to time management finds its origins in the industrial era, where businesses and factories, recognizing the correlation between time efficiency and economic productivity, transformed time into a commodity [53]. This shift was further propelled by Frederick Taylor's advocacy of scientific management, also known as 'Taylorism,' which was aimed at optimizing work in factories by determining the most efficient ways for employees to perform tasks [84]. The concept of lean project management, as presented in The Toyota Way [52], epitomizes current day's efforts to maximize efficiency within a capitalist society. Temporal artifacts emerged as a result, situated at the intersection of regularity, coordination, and standardization [39]. The widely accepted metaphor 'time is money' encapsulates the commodification of time, emphasizing its instrumental value. The act of ticking off items on a to-do list and transitioning from one 'now' to another 'now', often in a frenetic race through the day, is driven by the belief that more tasks accomplished equate to increased earnings. However, the pursuit of efficient time management, despite its ability to enhance productivity, does not inherently address or ensure a fulfilling life [40].

Timekeepers in the digital era, including computers, smart devices, online calendars, and scheduling tools, inherit inspiration from their physical predecessors and now shape the hybrid physical-digital lifestyles of individuals [43]. These digital time artifacts aid in time management by organizing activities into finite 15- or 30-minute blocks, each designated for a specific purpose and movable in a linear order [56]. Notifications on screens prompt individuals to shift their attention from one time slot to another, facilitating a seamless transition between work, leisure, and personal time. The ticking clock, once a standalone physical device, now echoes in technologies, orchestrating the rhythm of daily life as devices guide individuals through their day. Its presence in human life and human dependency on it is apparent only when the setup momentary fails—either as a system error or human mismanagement; otherwise, its presence goes unnoticed. Paraphrasing from Marshall McLuhan's view, "We shape our clocks and then our clocks shape us" [58].

### 4.2 Lost perspectives of multicultural time

Timekeeping devices serve as reflections of prevailing societal norms from that time period [39]. Historically, humans relied on natural time markers, relying on temporal concepts like the solar year, lunar month, and solar days based on Earth's celestial movements [3]. Sundials visually segmented solar days, representing temporal hours as they passed [39]. However, the need for a timekeeping device that was independent of natural cues arose in regions that experienced large variations in temporal hours year-round e.g., in Europe [51]. In medieval Europe, the sophistication of timekeeping instruments grew and a significant leap forward occurred with the development of mechanical clocks [39]. The church played a pivotal role in advancing clock technology, with monasteries and cathedrals housing intricate devices primarily used to indicate precise

prayer times [4]. By the late fourteenth century, the limitations of church bells for tracking working hours spurred innovations in mechanical clocks, eventually leading to wristwatches [61]. What began as a monastic practice gradually established a clock rhythm in European social life, and this narrative eventually spread globally through travel, trade, and colonization [14].

Cultural narratives and traditions intricately weave into temporal dimensions, providing unique insights into a society's collective consciousness; for instance, in India, time holds cultural significance in ancient texts and rituals [23]. Festivals, ceremonies, and rituals serve as cyclic markers, creating a shared temporal identity among community members. Additionally, cultural differences manifest in how people from various cultures communicate about time through language [9, 11, 49]. Cultures favoring linear time emphasize scheduling and punctuality, contrasting with those embracing variable time schedules as adaptable [59]. These cultural variations impact not only daily routines but also social interactions and expectations. The balance between past-oriented values (tradition) and future-oriented values (development) influences how individuals from different cultures perceive time in relation to space [16]. The cultural perspective on time forms a rich landscape resulting from historical influences, societal values, and collective imagination, shaping both individual experiences and broader social structures, thus contributing to diverse cultural identities.

Non-linear perspectives, drawn from cultural narratives and traditions, can offer alternative frameworks that prioritize adaptability and cyclic markers, disrupting conventional notions of clock time. It is time for societies to embrace the rich temporal dimensions and resist the rigid scheduling arising from clock dominance. Such a shift can not only shape individual experiences but can also challenge broader social structures and reshape cultural identities. Decolonizing design practices that involve the use of anachronism and timelessness are being suggested as powerful tools to challenge hegemonic futures [29]. More such approaches are needed to allow for the emergence of diverse temporal narratives in the global technological landscape dominated by the Western world. A global acknowledgment of diverse cultural perspectives is the need of the hour for fostering a dynamic dialogue that redefines our understanding of temporal dimensions and thus helps in designing temporally inclusive technologies.

### 4.3 Implications of dominant clock time

Temporal artifacts like digital clocks, smartphones, and time management tools, while undeniably convenient and essential for our organized and interconnected world, often prioritize the linear and measurable aspects of time, overshadowing the subtleties of the lived experience [36]. These artifacts rely on shared rhythms and temporal language to convey time effectively, creating challenges when these shared notions don't align with individuals, such as people with dementia depicted in Figure clockdrawing. For them, the clock may not communicate time, rather it may impose an externally synchronized rhythm onto their intuitive and subjective internal experience. Beyond dementia, the need for multiple temporalities has been spoken of for people with other neurological disturbances [33]. Similarly, 'crip time' advocates for 'a flexible standard for punctuality' to consider the extra time needed due

to disability [10] and Ellen Samuels provides different lenses to explore the concept of 'crip time' by connecting each lens with personal stories [76]. Such populations will largely benefit from approaches that break the rigid structure of external timekeeping.

Temporal rigidity extends beyond artifacts to environments like hospitals and care facilities, where clocks coordinate activities, create predictability, and establish routines. Despite employing ideas of person-centered care from Kitwood [48], the rigidity of medical environments makes it challenging to address individual subjective needs, resulting in tensions between subjective and objective expectations. In a way, the living environment of one person (person with dementia) is another person's (formal caregiver and medical staff) work environment. Caregivers, being part of post-industrial societies, often treat clock time as a commoditized entity, aligning with their work culture. Recognizing this is crucial for understanding the temporal dynamics in care settings and addressing clashes in subjective and objective temporal expectations.

In the context of technology development and caregiving, post-industrial societies treat clock time as a commoditized entity, impacting user-centered methodologies. The concept of "need for orientation" is often biasedly equated with "alignment to clock time," neglecting the nuanced temporal needs of individuals like that with dementia. The next section explores potential directions to creatively address this challenge.

## 5 IF NOT THE CLOCK, THEN WHAT?

Just as beauty is subjective and lies in the eyes of the beholder, time, too, is deeply personal. It is time for HCI research to explore time through technologies in more creative ways such that they integrate well with rich, diverse, and personal perspectives, considering different cultures and abilities. Communication of time and its passage can be thought of in varied and more intuitive ways such that it is inclusive of the internal, subjective, and imaginative dimensions of human temporality.

*Creative exploration of familiar artifacts* can be the first step towards re-expanding the notions of time again and making time a tangible entity through intuitive artifacts. The inherently abstract concept of time has been explored and debated through various metaphors, such as time as a river [55, 80]. Such metaphors of time that relate to the space we live in offer relatable perspectives on time passage. Indicators from surroundings offer a perspective that can ground individuals in the space they are a part of, be it through morning-time's gentle glow, night-time's comforting embrace, the warmth of summertime, and the slow changes of various seasons. Humans use their physical interactions with the world around them to gain a perspective of abstract ideas, such as time, that cannot be seen or touched. Thus, the use of familiar, experiential metaphors in combination with human senses can be used to effectively communicate time and its passage with someone like Lisa (Scenario 3.1.2 Flowing through time) and help her intuitively navigate through the flow of her lived lifetime. This can be through artifacts that act as symbolic indicators of time passage for different time granularities, e.g., an hourglass, a familiar artifact can be re-imagined to capture and communicate time not just for the time frame of an hour but for time-frames spanning mere moments to larger durations. The creative exploration of tangible artifacts to capture time and its

passage need not be limited to metaphors; it can also be expanded to include various ways in which people express time via language.

*Language can be used as a material for inspiration* when designing effective human-technology interactions that embrace indeterminateness in planned activities. Humans use language to express their experiences of time, capturing subjective nuances [9, 11, 49]. This linguistic flexibility goes beyond definitive structures laid by the clock offering insights into our collective negotiation with time, especially in cultural notions where language plays an integral role. The influence of language on time perception is evident, with English speakers visualizing time horizontally and Mandarin speakers associating it vertically [11]. Despite these variations, language can use activities and moments as narrative anchors, accommodating vagueness not captured by clocks. Language can associate activities with timeframes thus giving a temporal structure to activities in a day like rituals around waking up, breakfast, runs, etc. On a larger scale e.g., lifetime, language helps humans connect their experiences from specific moments to specific time periods— such as that time when I fell. Developing technologies that actively perceive differences and possess an awareness of the future in relation to the past is proposed as the first step for designing human-technology interactions aligned with human temporal dimensions [24]. This is very timely given recent AI advancements that have made it possible to interact with people on an individual level [92]. Exploring Large Language Models (LLMs) offers avenues for developing temporally-aware conversational interfaces. This could be of particular help when designing technology adaptable to the fluidity of David's time perception (Scenario 3.1.1 Navigating a temporal whirlwind). For such use cases that require technology to conduct temporally-aware conversations, LLM's versatility, accessibility, and ability to provide personalized interactions [30, 60] could be leveraged. However, visualizing the needs of people with dementia and delving into such creative and diverse perspectives that go beyond the norm of clock time can be challenging.

*Storytelling in de-familiarised context* offers a tool for breaking the norms and critically reflecting on the mainstream approaches in design [8]. For example, such an approach has been proposed as a valuable mechanism for prompting reflection on the taken-for-granted connections between well-being and a world of familiarity in a post-COVID world when confronting ableism [88]. Another example of creatively using de-familiarisation is Abbott's Flatland [1] where a square who is an inhabitant of a two-dimensional universe is introduced to the mysteries of three-dimensional space by a sphere. The narrative helps readers explore the new, yet unknown world by drawing on the cues from the surroundings, eventually making the initial abstractness logical through observed patterns. In this paper, we tried to demonstrate a similar approach for the context of dementia through the story of Alice (from the book Alice's Adventures in Wonderland [17]) at the start of the paper. The goal here was to bring forward the notions of time that are dominant in current technologies and how they do not align with the nuances of the lived experience of dementia as a concrete case. The metaphors from The Wonderland offer a new perspective that is still familiar and relatable for most readers while still offering a new perspective for re-evaluation.

In essence, the call for designers extends beyond creating yet another simplified clock to designing more creative timekeeping devices and intuitive human-technology interactions. It beckons a thoughtful exploration of human experiences, leveraging the use of language and metaphors, delving into the nuanced needs of specific user groups like those with dementia, and harnessing the potential of subjective experiences for developing temporally-aware technologies.

## 6 TOWARDS INCLUSIVE TEMPORALITIES IN TECHNOLOGY

Time, a fundamental aspect of the human experience, plays a pivotal role in shaping how we live and communicate. The prevalent norm of clock time in technology design overlooks the multi-dimensional and nuanced notions of time that shape human experiences and thus limit personalization posing risks for cognitively affected populations like dementia. Commoditized clock time is often unquestioned when integrated into digital technologies. But dementia, with its temporal complexities, serves as a compelling example illustrating the current mismatch between technology and care, leading to unintended consequences. The effectiveness of temporal artifacts relies on a shared sense of temporality between the users and the artifacts, beyond clock time, allowing people with dementia to trust technology better since it can be more attuned to them. We thus discussed scenarios that critically help examine the nuanced temporal needs of individuals vs. the often rigid temporal frameworks imposed by everyday technology. By highlighting temporal misfits, particularly in the context of dementia, this paper hopes to inspire future designs in which technology can better account for varied and nuanced temporal experiences of people. We challenge the current dominant norm of clock time in assistive technology design, urging the CHI community to broaden our notion of time to make space for inclusive designs that demonstrate diverse temporal experiences. We urge designers to come up with creative and original modalities of time that cater to people's intuitive exploration of time while prioritizing temporal flexibility based on individuals' changing contexts.

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